

LESSON 6.6

SOLVING REAL-WORLD PROBLEMS USING SIMILAR TRIANGLES



LESSON INTRODUCTION

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

LEARNING TARGETS

- I can solve real-world problems using similar triangles.

BENCHMARK COHERENCE

BUILDING ON

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

WORKING TOWARD

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

ESSENTIAL QUESTIONS

- How can you apply what you have learned about similar triangles to solve real-world problems?

LESSON NARRATIVE

In this lesson, students extend their learning from the previous lesson to include solving real-world problems with similar triangles. Initially, students develop an understanding of corresponding angles in similar triangles and their relationships. Next, students explore drawing diagrams, identifying corresponding sides, and using similar triangles to solve real-world problems.

COMPONENT SUMMARY

6.6.1 WARM-UP (~5 MIN.)

Reflect on visual representations in math

6.6.3 GUIDED INSTRUCTION (10 MIN.)

Solve real-world problem problems involving heights, shadows, and the sun

6.6.5 COLLABORATION (10 MIN.)

Practice solving real-world problems involving similar triangles

6.6.7 WRAP-UP (~5 MIN.)

Demonstrate mastery of lesson learning targets

6.6.2 EXPLORATION (10 MIN.)

Explore corresponding angles in similar triangles

6.6.4 COLLABORATION (5-10 MIN.)

Identify and correct errors in similar triangle problems

6.6.6 YOUR TURN! (5 MIN.)

Additional practice solving real world problems involving similar triangles

RESOURCES

GLOSSARY

Key Terms:

- Corresponding angles

Additional Terms:

- Similar
- Corresponding sides
- Constant of proportionality

MATERIALS

- Patty paper
- (Optional) Calculators
- (Optional) Highlighters

ADDITIONAL PREPARATION

- Warm-Up 6.6.1 contains a video to accompany the question prompts.
- Consider creating an anchor chart with visual components and definitions of vocabulary.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle visualizing the context of the problem and representing it in a diagram.
- Students may not correctly identify corresponding sides.
- Students may incorrectly calculate the constant of proportionality.

THINGS TO CONSIDER

- Consider enlarging diagrams and images to accommodate for visually impaired students.
- Consider allowing students use calculators so they can focus on the how to set up, solve, and interpret the application problems.
- If time is an issue, then consider splitting the lesson into two days where 6.6.5 Collaboration is used as a gallery walk, or assigning 6.6.5 Collaboration and/or 6.6.6 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 6.6.2 Exploration, students explore how angles of similar triangles are related. After students explore the similarities and differences, encourage students to think about and then share what they notice and wonder about similar triangles and their corresponding angles.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 World Contexts

In 6.6.4 Collaboration, students create concrete and abstract models to solve real-world problems involving similar triangles. Students discuss their thinking and strategies with their partners. During the whole-class debriefing, a variety of students share their strategies. These discussions will help students build confidence and understanding.

DIFFERENTIATE INSTRUCTION

In 6.6.6 Your Turn!, students may have trouble finding corresponding sides as the triangles have different orientations. If they are struggling with this, consider having them think back to the strategies from Lesson 5.

- Highlight the sides that are corresponding using different colors.
- Look for the “short,” “medium,” and “long” sides.

Additional differentiation opportunities are denoted throughout the Lesson Guide.

6.6.1 WARM-UP: BRAINS THINK ABOUT MATH VISUALLY

TEACHER MOVES

Play the “Brains Think About Math Visually” video for the entire class. After students watch the video (3:11 minutes), prompt them to reflect on the video by answering the two questions independently.

The video features Dr. Jo Boaler, the author of “Limitless Mind: Learn, Lead, and Live Without Barriers.” She has done extensive research on how different teaching approaches impact students’ learning, how to teach mathematics for a “growth mindset,” and how equity is promoted in mathematics classrooms. Students are often shamed into thinking that using visuals such as counting on their fingers indicates their inability to do mathematics. Use this opportunity to build students up.

Consider having students share their examples and offer positive feedback. Students will benefit from hearing a variety of examples. Ask students to share examples from their work in math in Kindergarten (i.e. counting on their fingers) through 5th grade (i.e. images of geometric shapes when finding the volume).



6.6.1 Warm-Up: Brains Think About Math Visually

1. Explain a time in math class when you used a visual representation of a problem to help solve it.

Answers will vary.

2. How did that visual representation help you?

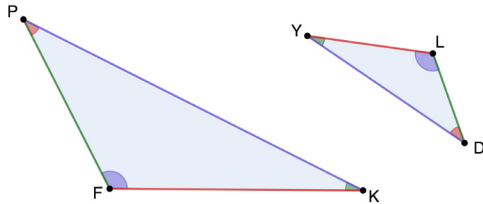
Answers will vary.



6.6.2 Exploration: Angles in Similar Triangles

The previous lesson introduced similar triangles and the relationship between their corresponding sides. The relative side lengths of the triangles in those situations were used to identify corresponding sides. In some cases, it may be more helpful to use angle measures to help identify corresponding sides.

Triangle *PFK* and triangle *DLY* are similar.



Line segment *PK* corresponds to line segment *DY*.
Line segment *PF* corresponds to line segment *DL*.
Line segment *FK* corresponds to line segment *LY*.

1. Mark the figures to show which sides correspond, based on the given information, using colors or symbols.
See image.

2. Identify the angles opposite each line segment.

Triangle <i>PFK</i>	
Line Segment	Opposite Angle
<i>FK</i>	$\angle P$
<i>KP</i>	$\angle F$
<i>PF</i>	$\angle K$

Triangle <i>DLY</i>	
Line Segment	Opposite Angle
<i>LY</i>	$\angle D$
<i>YD</i>	$\angle L$
<i>DL</i>	$\angle Y$

3. Angles opposite corresponding sides are called **corresponding angles**. Use colors or markings to show which angles correspond to each other on the triangles.
See image.
4. Complete the statement by identifying corresponding angle pairs in triangles *PFK* and *DLY*.

 $\angle PFK$ corresponds with $\angle DLY$.
 $\angle FKP$ corresponds with $\angle LYD$.
 $\angle KPF$ corresponds with $\angle YDL$.
5. Consider the size of each pair of corresponding angles. With your partner, write a hypothesis regarding the relationship between each angle pair.
Answers will vary.
6. Trace a copy of triangle *PFK* on patty paper. Be sure to label each vertex.
7. Place the copy of triangle *PFK* on top of triangle *DLY* so that angle *PFK* overlaps angle *DLY*. What do you notice?
Answers will vary. Possible answers: the angles are the same size, the legs of the triangles are not the same size eg: triangle PFK has longer legs/side lengths or triangle DLY has shorter legs/ side lengths

6.6.2 EXPLORATION: ANGLES IN SIMILAR TRIANGLES

TEACHER MOVES

The Exploration builds on Lesson 5. It further lays the foundation of the properties of similar triangles by showing students that the corresponding angles of similar triangles are congruent.

TEACHER MOVES

It is not necessary for students to use traditional markings for this. Teachers can choose to show students how to use hash marks to identify corresponding sides, but students could just as easily use highlighters or markers of different colors to show which sides correspond. The purpose here is simply to visually show the information that is given above so they can use it on the next part of the activity more easily.

DIFFERENTIATE INSTRUCTION

Consider creating and displaying an Anchor Chart with the definition, examples, and visuals of corresponding angles for students to reference.

TEACHER MOVES

Traditional markings are not required, but teachers can choose to show students these markings or have students use colors to mark the corresponding angles.

6.6.2 EXPLORATION: ANGLES IN SIMILAR TRIANGLES (CONTINUED)

TEACHER MOVES

Notice and Wonder

Students explore the relationship among corresponding angles in similar triangles. Encourage students to reflect and then share what they notice and wonder about similar triangles and their corresponding angles.

8. Repeat the process for each pair of corresponding angles.
9. Discuss with your partner whether your observations support or disprove your hypothesis about the relationship between each angle pair. Then, summarize your discussion.
Answers will vary. Students should notice that angles to similar triangles are the same size/degree measure.



In similar triangles, **corresponding angles** have equal measures and are opposite corresponding sides.

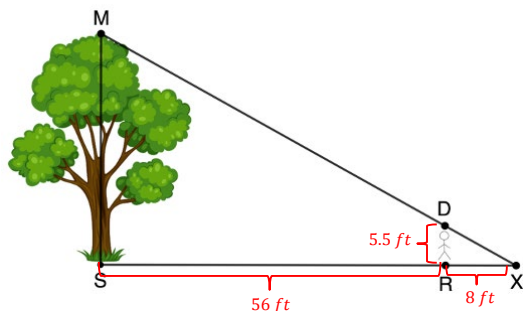


6.6.3 Guided Instruction: Using Similar Triangles to Solve Real-World Problems

The largest native tree in Florida is a bald cypress located in Hamilton County. It's 101 feet (ft.) tall. In a local park, there is a very tall tree. Ryan wants to find out if the tree in the local park is close to the height of the bald cypress.

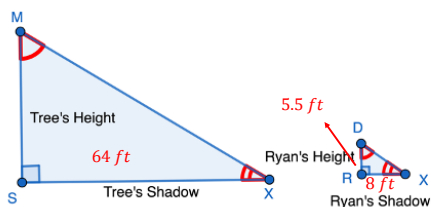
When two objects are near one another, the sun hits those objects at approximately the same angle. Ryan walks toward the tree until his shadow and the tree's shadow end at the same point. He is exactly 5.5 ft. tall and casts a shadow of 8 ft. When the shadows are at the same point, Ryan is standing 56 ft. from the base of the tree.

1. Label the diagram below with the given information.



2. How long is the tree's shadow?
 $56\text{ ft} + 8\text{ ft} = 64\text{ ft}$
The tree's shadow is 64 feet long.

3. The diagram below shows two similar triangles that can be used to represent the situation.



- a. Mark the corresponding angles and label the sides with the known measurements.
 - b. Determine the constant of proportionality, relating the length of the tree's shadow to the length of Ryan's shadow.

$$\frac{\text{Tree's Shadow}}{\text{Ryan's Shadow}} = \frac{64\text{ ft}}{8\text{ ft}} = 8\text{ ft}$$
 - c. Use the constant of proportionality to determine the height of the tree.

$$\frac{64\text{ ft}}{8\text{ ft}} = \frac{x}{5.5\text{ ft}} \quad x = 44\text{ ft} \quad \text{OR} \quad 8 \times 5.5 = 44\text{ ft}$$
4. Explain whether or not the height of the tree in the park is close to the height of the bald cypress in Hamilton County.
The height of the tree in the park is about 57 feet shorter than the height of the bald cypress in Hamilton County.

6.6.3 GUIDED INSTRUCTION: USING SIMILAR TRIANGLES TO SOLVE REAL-WORLD PROBLEMS

TEACHER MOVES

Consider asking students to highlight or underline the key information in the problem. Then, ask students to identify where the information was used in the diagram. This will help students when they need to create their own diagrams in future problems.

TEACHER MOVES

Students may not see two triangles because the triangles overlap. Consider pausing to draw the two triangles separately. Then, have the students highlight one of the triangles in their workbook. Consider scaffolding the process of finding the length of the tree's shadow using these questions:

- What are the corresponding sides?
- What are the corresponding angles?
- How can we find the length of the tree's shadow?

This will help the students make sense of the diagram before question 1a, which shows the two separate triangles.

DIFFERENTIATE INSTRUCTION

Consider printing or drawing larger diagrams to ensure all students, including those with visual impairments, can clearly read the information in the diagram.

TEACHER MOVES

Consider providing students with the following scaffolding to help them set up their ratio:

$$\frac{\text{Tree's Shadow}}{\text{Ryan's Shadow}}$$

6.6.4 COLLABORATION: ERROR ANALYSIS

ENRICHMENT

How do you determine your height? What are the corresponding sides and angles in the triangles?

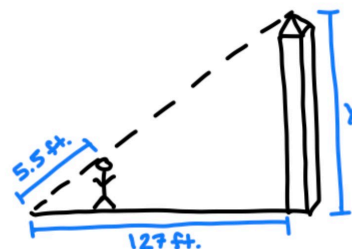


6.6.4 Collaboration: Error Analysis

Erika and Michael were working together on the following problem:

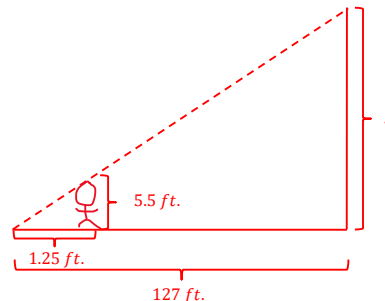
A man is standing near the Washington Monument. He is 5.5 feet (ft.) tall and casts a shadow that is 1.25 ft. long. The shadow of the Washington Monument is 127 ft. long. What is the height of the Washington Monument?

1. They drew the following visual representation to help solve the problem.



- a. There is an error in their diagram. Describe the mistake they made.
The man's height in the diagram is not labeled correctly. This diagram has the man's head to the end of the shadow labeled 5.5 feet, when it should be from the man's head to the ground/his feet.

- b. Draw a correct diagram. Label the sides with the known measurements.



- c. Find the height of the Washington Monument.

$$\frac{\text{Length of the monument's shadow}}{\text{Length of the man's shadow}} = \frac{127 \text{ ft}}{1.25 \text{ ft}} = 101.6 \text{ ft}$$

101.6 is the constant of proportionality

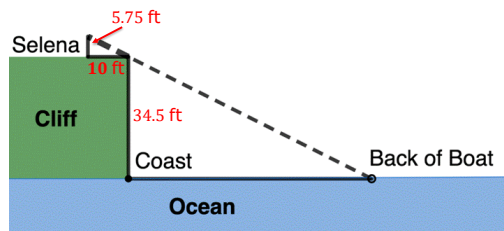
$$101.6 \text{ ft} \times 5.5 \text{ ft} = 558.8 \text{ ft}$$

Multiply the constant of proportionality (101.6 ft) by the man's height (5.5 ft). The monument is 558.8 ft tall.



6.6.5 Collaboration: Using Similar Triangles to Solve Real-World Problems

- While hiking, Selena comes to a cliff overlooking the ocean. She notices a ship in the distance and wonders how far the ship is from the coast. She knows that the cliff is 34.5 feet (ft.) above the water. If she stands 10 ft. from the edge of the cliff, she can visually align the top of the cliff with the water at the back of the ship. Her eye level is $5\frac{3}{4}$ ft. above the ground.

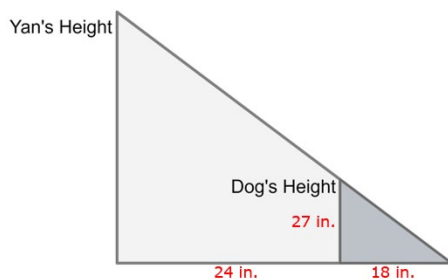


- Label the similar triangles with the given information.
- Find the distance from the coast to the ship.

$$\frac{\text{Height of the cliff}}{\text{Height of Selena}} = \frac{34.5 \text{ ft}}{5.75 \text{ ft}} = 6$$

6 is the constant of proportionality. The distance from the coast to ship is $10 \times 6 = 60$ ft. 10 ft is the distance that she is standing from the edge of the cliff and 6 is the constant of proportionality.

- Yan is walking his dog. At a certain point in their walk, he notices that their shadows end at the same spot. Yan's dog is 27 in. tall. His dog is walking 24 in. in front of him. The dog's shadow is 18 in. long.



- Label the similar triangles with the given information.
- Find Yan's height.

$$\text{The length of Yan's shadow: } 24 \text{ in.} + 18 \text{ in.} = 42 \text{ in.}$$

$$\frac{\text{Yan's shadow length}}{\text{The dog's shadow length}} = \frac{42 \text{ in.}}{18 \text{ in.}} = \frac{7}{3}$$

$\frac{7}{3}$ is the constant of proportionality. Multiply the dog's height by the constant of proportionality to solve for Yan's height.

$$27 \text{ in.} \times \frac{7}{3} = 63 \text{ in.}$$

Yan is 63 inches tall.

6.6.5 COLLABORATION: USING SIMILAR TRIANGLES TO SOLVE REAL-WORLD PROBLEMS

TEACHER MOVES

This activity can be structured as a jigsaw and then debriefed with a Gallery Walk. Consider assigning different problems to different groups. Then, ask each group to display their work on a poster. Finally, have students present their posters as students in other groups circulate to view the other groups' posters. Have students carry a notebook or clipboard to take notes while visiting other groups.

DIFFERENTIATE INSTRUCTION

There is a lot of reading in the word problems in this Collaboration. Consider strategically pairing students to provide peer support during collaboration. For example, ask a group member to read the problem aloud. Ask students to discuss the context to ensure that all students understand the problem. Remind students to highlight key information in the problem and share what they highlighted with their group members.

ENRICHMENT

- What information are we given?
- What are the corresponding sides and angles?
- Is your answer reasonable/does it make sense in the context of the problem?

DIFFERENTIATE INSTRUCTION

Consider allowing students to use calculators to determine the constant of proportionality. Ask students to write the ratio of Yan's to the dog's shadow lengths before using the calculators.

6.6.6 YOUR TURN!

DIFFERENTIATE INSTRUCTION

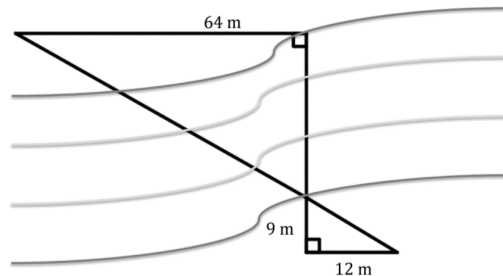
If students are struggling with identifying the corresponding sides, consider having them think back to the strategies from Lesson 5.

- Highlight the corresponding sides
- Look for the “short,” “medium,” and “long” sides
- Cut out the triangles and rotate them so they have the same orientation
- Use patty paper to trace a triangle and rotate it so the triangles have the same orientation



6.6.6 Your Turn!

1. A surveyor is measuring the width of a river for a future bridge. He is unable to measure across water, so he uses similar triangles and takes a series of measurements on land. The diagram shows his measurements.



Use the surveyor's measurements to determine the width of the river.

$$\frac{64 \text{ m}}{12 \text{ m}} = \frac{16}{3}$$

$\frac{16}{3}$ is the constant of proportionality. $\frac{16}{3}$ multiplied by 9 m is 48 m. The river is 48 meters wide.



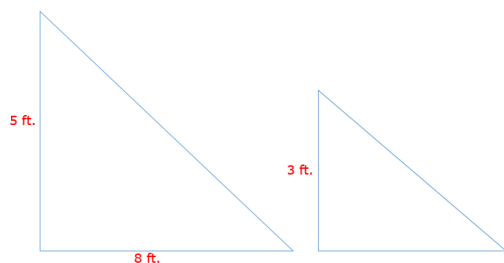
6.6.7 Wrap-Up: Ticket Out the Door

- Shanice and her younger brother are walking home from school. Shanice notices that her shadow is 8 feet (ft.) long. Shanice is 5 ft. tall and her brother is 3 ft. tall.

Determine the length of her brother's shadow.

Shanice:

Shanice's brother:



$$\frac{\text{Shanice's brother's height}}{\text{Shanice's height}} = \frac{3 \text{ ft}}{5 \text{ ft}} = 0.6$$

0.6 is the constant of proportionality. Multiply Shanice's shadow length by the constant of proportionality to get Shanice's brother's shadow length.

$8 \text{ ft}(0.6) = 4.8 \text{ ft}$. Shanice's brother's shadow is 4.8 ft long.

6.6.7 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Consider using student responses from prior self-reflections with this Ticket Out the Door to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.